

Sentiment Analysis Platform

Local Architecture, Validation Processing, and Evaluation Roadmap

System Framework Milestone Review

Team Details :

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Topic: Social media sentiment analysis

Group No. 80

| System Overview & Interaction Flow

The application follows a strictly decoupled client-server pattern mapping asynchronous requests directly to functional python layers:

- **Decoupled UI Communications:** Dynamic form capture triggers backend processing vectors without page resets.
- **Python API Layer:** Processes multi-tiered evaluation steps using lightweight localized endpoint routes.
- **State Tracking Array:** Maintains interaction historical indexes within local dynamic storage frameworks.

| Local Repository Directory Layout

Workspace organized across distinct module boundaries to ensure isolation of pipeline assets and report logs:

```
social-media-sentiment-analysis/  
├── data/      --> Processed CSV logs (cleaned_social_media_data.csv)  
├── src/       --> Core logic script engine routines  
├── reports/   --> Metric evaluation frameworks and logs  
└── deployment/ --> Environment binding profiles (http://127.0.0.1:5000/)
```

| Data Cleaning Execution Run Log

Automated verification logs generated via data cleansing routine executions in
src/main.py:

```
> python src/main.py  
hello social media sentiment  
--- Starting Data Cleaning Phase ---  
Original Dataset Shape: (5, 4)  
Shape after removing missing text: (4, 4)  
Shape after removing duplicate profiles: (4, 4)  
Success! Cleaned dataset exported smoothly to: data/cleaned_social_media_data.csv
```

| Functional Backend Routing Endpoints

POST /api/chat

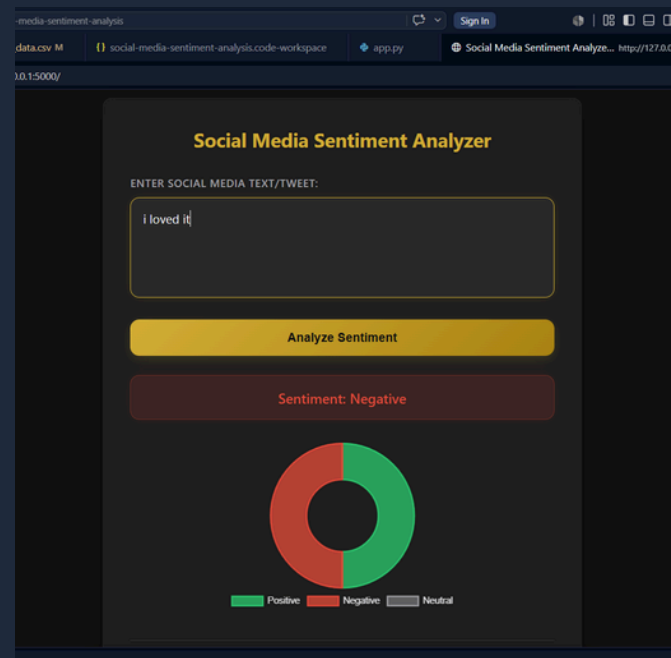
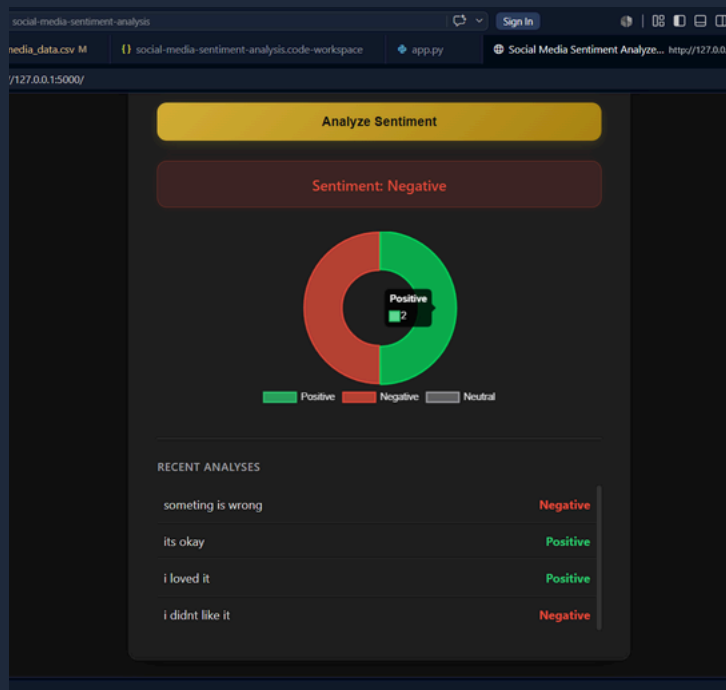
Consumes client payload inputs, routes structures to lexicon parsing blocks, and flushes directional calculations across standard scoring tiers.

GET /api/history

References in-memory structural lists to return full request interaction footprints bundled with processing timestamps.

Empiric System Validation Profiles

Input Text Sample Under Evaluation	Category Matrix	Expected Behavior Profile
"The new dashboard interface is incredibly smooth and fast, but the login system keeps giving me errors."	Mixed Metric	Splits calculations across opposing dimensions.
"The application update was successfully downloaded and installed on the system."	Factual Neutral	Locks score calculation directly to zero index markers.
"This cafe has two separate indoor dining rooms and an outdoor seating area."	Objective Observation	Detects no emotional polarity indicators.



| Session Tracking Logs Payload Validation

In-memory tracking snapshot retrieved directly from state storage endpoints:

```
[
  {
    "text": "The new dashboard interface is incredibly smooth and fast, but the login system keeps giving me errors.",
    "sentiment": "Mixed",
    "timestamp": "12:04:10"
  },
  {
    "text": "The application update was successfully downloaded and installed on the system.",
    "sentiment": "Neutral",
    "timestamp": "12:04:18"
  }
]
```

| Milestone Handoff Status Checklist

- **Pipeline Operations:** `src/main.py` isolates records and outputs standardized schema assets.
- **API Execution:** Endpoints map asynchronous validation scoring logic correctly.
- **Layout Compliance:** Directory tree nodes built out cleanly with indexing markdown documentation.

Local Architecture Evaluation Deliverables Complete

THANKYOU